FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2023

CONCEPT RECAPITULATION TEST – I

PAPER –2

TEST DATE: 26-04-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 – 04, 19 – 22, 37 – 40)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 23 – 28, 41 – 46)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 18, 29 – 36, 47 – 54)*: This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

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Physics

PART – I

Section – A (Maximum Marks: 12)

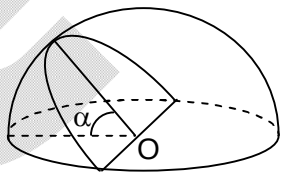
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. Three small balls of the same mass, white (w), green (g), and blue (b), are fixed by weightless rods at the vertices of the equilateral triangle with side l. The system of balls is placed on a smooth horizontal surface and set in rotation about the centre of mass with period T. At a certain instant, the blue ball tears away from the frame.
Determine the distance L between the blue and the green ball after the time T.

(A) $\sqrt{\frac{2}{3}+\left(\pi\sqrt{3}-\frac{1}{2}\right)^2}$ (B) $\sqrt{\frac{4}{3}+\left(\pi\sqrt{3}-\frac{1}{2}\right)^2}$

(C) $\sqrt{\frac{4}{3}+\left(\pi\sqrt{3}-1\right)^2}$ (D) $\sqrt{\frac{3}{4}+\left(\pi\sqrt{3}-\frac{1}{2}\right)^2}$

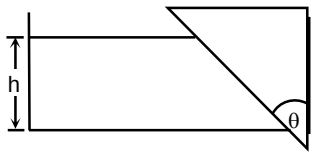
2. The strength of the electric field produced by charges uniformly distributed over the surface of a hemisphere at its centre O is E_0. A part of the surface is isolated from this hemisphere by two planes passing through the same diameter and forming an angle α with each other. Determine the electric field strength E produced at the same point O by the charges located on the isolated surface (on the "mericarp").



(A) $\frac{E_0}{2}\sin\frac{\alpha}{2}$ (B) $\frac{2E_0}{3}\sin\frac{\alpha}{2}$

(C) $E_0\sin\frac{\alpha}{2}$ (D) $2E_0\sin\frac{\alpha}{2}$

3. A slit is cut along the right bottom edge of a rectangular tank. The slit is closed by a wooden wedge of mass m and apex angle θ as shown in the figure. The vertical plane surface of the wedge is in contact with the right vertical wall of the tank. Coefficient of static friction between these surfaces in contact is μ. To what maximum height, can water be filled in the tank without any leakage through the slit? The width of tank is b and density of water is ρ.



(A) $\sqrt{\frac{2m}{\rho b(\tan\theta-\mu)}}$ (B) $\sqrt{\frac{4m}{\rho b(\tan\theta-\mu)}}$

(C) $\sqrt{\frac{2m}{\rho b(\sin\theta-\mu\cos\theta)}}$ (D) $\sqrt{\frac{2m\cos\theta}{\rho b(\tan\theta-\mu\cos\theta)}}$

4. Radii of a slightly tapered cylindrical wire of length L at its ends are a and b. It is stretched by two forces each of magnitude F applied at its ends. The forces are uniformly distributed over the end faces. If Young's modulus of material of the wire is denoted by Y, extension produced is

(A) $\frac{4FL}{\pi(a+b)^2Y}$ (B) $\frac{FL}{\pi(a^2+b^2)Y}$

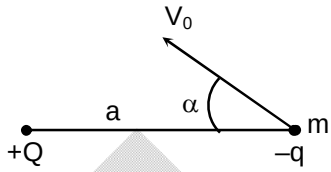
(C) $\frac{FL}{\pi(a^2-b^2)Y}$ (D) $\frac{FL}{\pi abY}$

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Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A charged particle of charge –q and mass m is projected with initial velocity $v_0 = \sqrt{\frac{qQ}{4\pi\varepsilon_0 am}}$ from a distance a from the fixed charge +Q at an angle of $\alpha = 60°$ with the line joining the charges as shown in the figure. It is found that the particle moves on an elliptical orbit with the charge +Q at one of its foci.



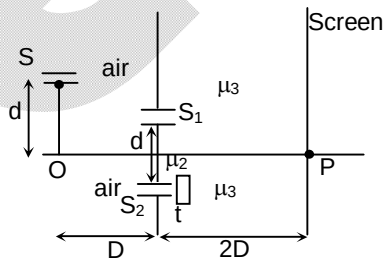
(A) The length of the semi-major axis of the ellipse equals to a

(B) The area of the ellipse orbit equals to $\frac{\sqrt{3}}{2}\pi a^2$

(C) The time period of one revolution will be $\sqrt{\frac{16\pi^3\varepsilon_0 ma^3}{qQ}}$

(D) The time period of one revolution will be $\sqrt{\frac{8\pi^3\varepsilon_0 ma^3}{qQ}}$

6. Consider the situation shown in the figure. Two slits S_1 and S_2 are placed symmetrically about the line *OP* which is perpendicular to screen and bisector to line joining the slits. The space between screen and slits is filled with a liquid of refractive index μ_3. A plate of thickness *t* and refractive index μ_2 is placed in front of one of the slit. A source *S* is placed above *OP* at a distance *d* in front of slit.



(Given that *D* = 1m, *d* = 2 mm, *t* = 6× 10^{-6}m., μ_2 = 1.2, $\mu_3 = 1.8$, Choose the correct alternatives

(A) Position of central maxima from point *P* is 2 mm.
(B) Position of central maxima from point *P* is 1 mm.
(C) If slab is removed, the central maxima shifts by a distance of 2 mm.
(D) If slab is removed, the central maxima shifts by a distance of 1 mm.

7. Two point masses, each of mass M are kept at rest at points A and B respectively. A third point m is released from infinity with a negligible speed, so that it can move along y-axis under the influence of mutual gravitational attraction on it due to point masses kept at A and B respectively as shown in the Figure -1. Figure –2 represents the potential energy of system (includes m, M at A and M at B) with position of m at y-axis. (Neglect any other forces other than gravity) (given Gm^2/d = 12 Joule(m = 6 kg). Choose the correct option(s)

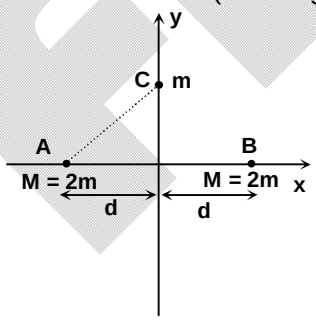


Figure -1

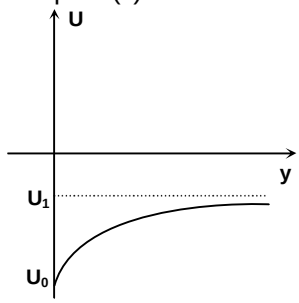
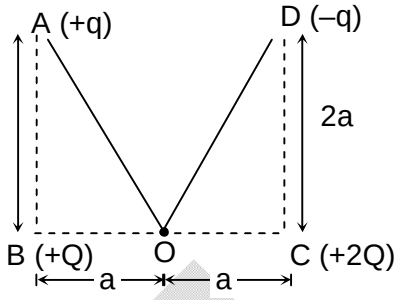


Figure -2

(A) Point mass m will perform periodic motion
(B) U_1 = –24 Joule
(C) Maximum speed of particle is 24 m/s
(D) Maximum speed of particle is 4 m/s

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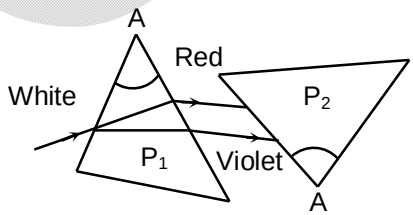
8. A ball at O is in equilibrium as it is attached with two strings AO and DO which are tied at A and D. AO = DO = $a\sqrt{5}$. The charges at A, B, C and D are +q, +Q, +2Q and –q respectively. Find the correct options. Find the correct options. The ball at O is positively charged :



(A) The ball O cannot remain in equilibrium
(B) If the charge at C is +Q, the ball will remain in equilibrium
(C) The ball O will remain in equilibrium
(D) If the charges at A and D and charges at B and C are interchanged, the ball will remain in equilibrium.

9. A metal cylinder of mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C. The cylinder temperature rises to 25°C in 5 min and finally becomes constant at 45°C. Assuming that rate of heat loss is proportional to the excess temperature over the surroundings, which of the following statements is/are correct.
(A) the rate of loss of heat of the cylinder to surrounding at 20°C is 2W.
(B) the rate of loss of heat of the cylinder to surrounding at 45°C is 12W.
(C) specific heat capacity of metal is $\frac{240}{\ell n(3/2)} Jkg^{-1\circ}C^{-1}$.
(D) specific heat capacity of metal is $\frac{240}{\ell n2} Jkg^{-1\circ}C^{-1}$.

10. P_1 and P_2 are identical prisms arranged as shown in figure. A ray of white light incident on face of P_1 undergoes dispersion and falls on one face of P_2. The facing surfaces of the prisms are parallel to each other and separation between them is negligible. Then,



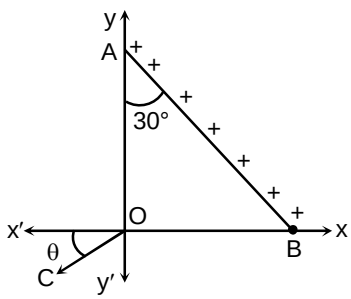
(A) Light emerging from P_2 will be white
(B) In the light emerging from P_2 dispersion will be greater
(C) the direction of light emerging from P_2 will be parallel to the direction of ray incident on P_1
(D) the ray emerging from P_2 will be white even it prisms P_1 and P_2 have identical geometry but different material

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

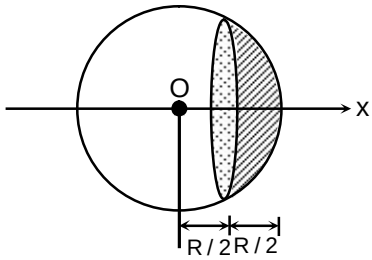
11. A long glass capillary tube of radius r is paced horizontally and filled with water (angle of contact for water – glass = 0^0) If the tube is made vertical, then the length of water column that remains in the capillary is $\frac{\eta T}{r\rho g}$ where 'T' is the surface tension of water, ρ is density water . Find η

12. A non-conducting rod of AB, having uniformly distributed positive charge of linear charge density λ is kept in x-y plane. The rod AB is inclined at an angle 30° with positive y-axis. The magnitude of electrostatic field at origin due to rod AB is E_0 N/C and its direction is along line OC. If line OC makes an angle θ = (10a + b) degree (where a and b are the positive integer less than 9) with negative x-axis, as shown in the figure, calculate the value of (a + b) [OB = 2m and $\lambda = 10^{-3}$ C/m]



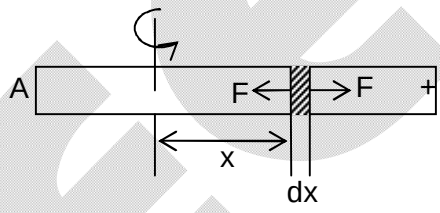
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13. The moment of inertia about X-axis of the solid spherical segment of mass M is $\frac{53MR^2}{40X}$. Then, the value of ρ is



14. The height of the point vertically above the earth's surface at which the acceleration due to gravity becomes 1% of its value at the surface is n R, where *R* is the radius of the earth. Find the value of n.

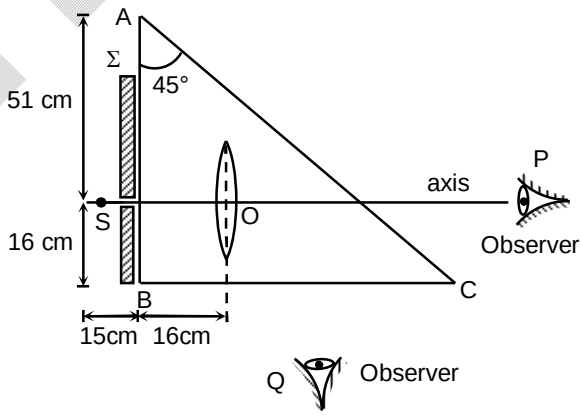
15. A horizontally oriented copper rod of length l is rotated about a vertical axis passing through its middle. Frequency at which the rod ruptures when rotated at $\frac{1}{2\pi}\sqrt{\frac{X\sigma}{\rho l^2}}$. Breaking or breaking force of copper takes as σ and density of copper ρ. Find X.



16. A non-viscous liquid of constant density 10^3 kg/m^3 flows in streamline motion along a vertical tube *PQ* of variable cross-section. Height of *P* and *Q* are 2m and 2.5m respectively. Area of tube at *Q* = 3 time area of tube at *P*. The work done per unit volume by pressure as liquid flows from *P* to *Q* is n $\times$ 1000 J/m^3. Find the value of n. Speed of liquid at *P* is 3 m/s (g = 10 m/s^2).

17. A radioactive sample contains two different types of radioactive nuclei; A-with half-life 5.0 days and B-with half-life 30.0 days. Initially the decay rate of the A type nuclei is 64 times that of B type of nuclei. Their decay rates will be equal when time is 9n days. Find the value of n.

18. An isosceles right angled triangular glass (μ = 1.6) prism has a cavity inside it in the shape of a thin convex lens whose both surfaces have radius of curvature equal to 20 cm. The cavity has been filled with a transparent liquid of refractive index 2.4. S is a point source and Σ is an opaque sheet having a small hole such that the source and the hole both lie on the principal axis of the lens. The small hole in the opaque sheet is just to ensure that only paraxial rays are incident on the optical system. However, the size of hole is large enough to neglect diffraction effects. Where is the image located?



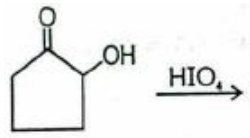
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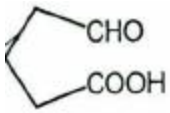
Chemistry

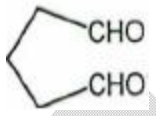
PART – II

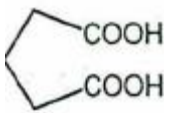
Section – A (Maximum Marks: 12)

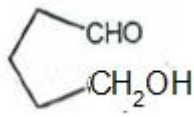
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

19.  Identify product

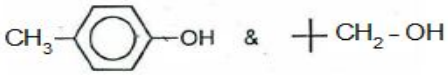
(A) 

(B) 

(C) 

(D) 

20. Compressibility factor for a gas (molar mass = 44 g/mol.) at 400 K and 71 bar is 0.869. Volume of 1 mole of gas in the given condition is
(A) 0.41 L (B) 19.5 L
(C) 22.4 L (D) 42.14 L

21. Which of regent listed here would serve as the basis for a simple chemical test to distinguish between

(A) $AgNO_3 + NH_4OH$ (B) Cold conc. H_2SO_4
(C) $NaOH/H_2O$ (D) Dilute HCl

22. Which element can form greatest number of Fluoride
(A) Hydrogen (B) Chlorine
(C) Bromine (D) Iodine

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. Milk of lime + $Cl_2(g) \xrightarrow{\text{Warm}} 'X' + 'Y'$
If oxidation state of chlorine in product 'Y' is positive, then select correct statement(s) for products 'X' and 'Y' of above given reaction.
(A) Both 'X' and 'Y' react with each other in the presence of excess of water
(B) The anionic part of 'Y' acts oxidizing agent
(C) 'X' is used as dehydrating agent
(D) When X and Y react together Cl_2 is liberated

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24. Select the correct option(s)

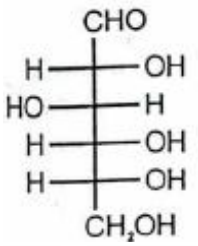
(A) The spin quantum number can have only two values

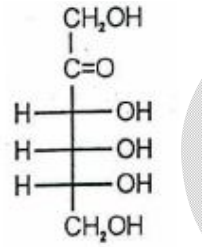
(B) Spin quantum number is obtained by solving Schrödinger wave equation.

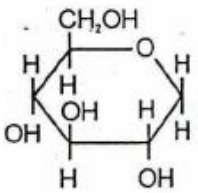
(C) Only one orbital is possible for quantum number $n = 3, \ell = 2, m = +2$.

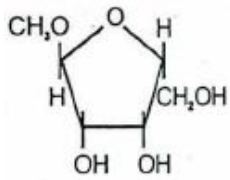
(D) The orbital angular momentum for all d-subshell electrons is $\frac{h}{2\pi}\sqrt{6}$

25. Which carbohydrate is/are reducing sugar?

(A) 

(B) 

(C) 

(D) 

26. Select incorrect statement(s)

(A) In rock salt (NaCl) structure $\frac{r_+}{r_-} < 0.414$

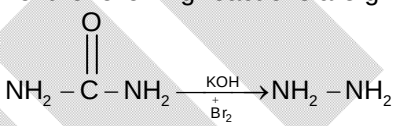
(B) In antifluorite structure anion forms fcc lattice

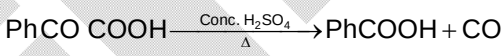
(C) In sphalerite structure (ZnS) coordination number of anion is 6

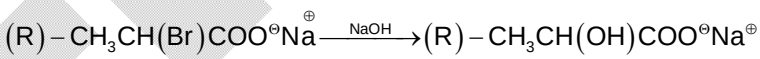
(D) In cesium chloride, $r_+ + r_- = \frac{\sqrt{3}a}{2}$

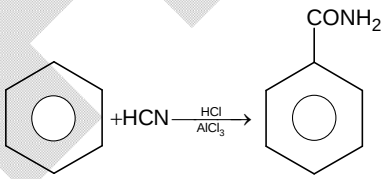
(Where a = Edge length, r_+ and r_- are ionic radius of cations and anion respectively)

27. Which of the following reactions are giving correct results?

(A) 

(B) 

(C) 

(D) 

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28. For given reactions

$Mg + NH_3(g) \xrightarrow{900^oC} 'X' + H_2(gas)$

$'X' + H_2O \longrightarrow 'Y' \downarrow + Z(gas)$

Select correct statement(s) for products 'X', Y' and 'Z'

(A) Precipitate 'Y' dissolves in presence of NH_4Cl

(B) Gas 'Z' turns $CuSO_4$ solution into deep blue solution

(C) Compound 'X' is nitrite of magnesium

(D) Gas 'Z' turns $Hg_2(NO_3)_2$ paper black

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. The equilibrium constant for the reaction: $2x(s) + y(g) \rightleftharpoons 2z(g), K_c = \frac{9}{16}$. If 4 moles of x and 3 moles of z are present in 4 L vessel at equilibrium what would be the moles of y at equilibrium

30. 80 g of NaOH is added to 1 L of 2 M $NiCl_2$ solution. Calculate the pH of final solution.
Given: K_{sp} of $Ni(OH)_2 = 10^{-16}$ at 25ºC

31. Find out total number of precipitates(s) that are soluble in hot dil HNO_3.
$HgS, Ag_2S, PbCrO_4, CaC_2O_4, FePO_4, CuS, BaSO_4, PbI_2, Ag_2S_2O_3$

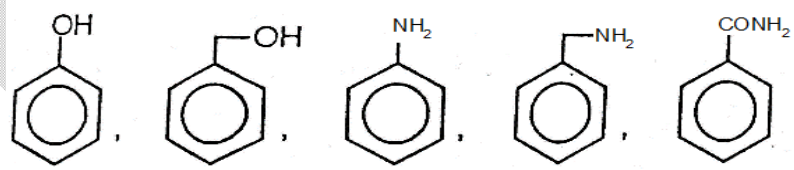
32. 0.5 mol of an ideal gas $(C_{v,m} = 2R)$ is expanded from $(400K, 2L)$ to 7L adiabatically against a constant external pressure of 0.821 atm. Calculate $\Delta S_{system} + \Delta S_{surr}$ $(\text{in cal } K^{-1})$.
(Rounded to nearest integer value) (Given: ln2 = 0.7; ln7 = 1.95)

33. For given reactions: $SO_2(g)$ —NaOH solution→ (X); $SO_2(g)$ —$H_2S(g)$→ (Y)

and $'X' + 'Y' \xrightarrow{warm} 'Z'$

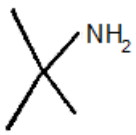
Find out algebraic sum of oxidation states of sulphur atoms present in a molecule of compound 'Z'

34. Which of the following compounds will give either of reminer-tiemann reaction or isocyanide test.

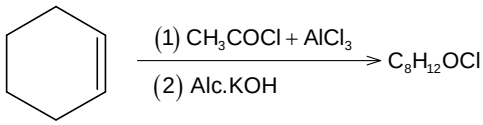


p-cresol, picric acid,

,



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35. 

How many stereoisomers does the product have

36. Find total number of mixed oxides among given oxides
Pb_2O_3, Fe_2O_3, ClO_2, P_4O_8, GaO, Fe_3O_4, Pb_3O_4, NO_2, SiO_2

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Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. If 2a + 2b + 3c = 1/5, and a, b, c > 0 then maximum value of constant term is expansion of $\left(ab\sqrt{x}+\frac{c}{\sqrt[3]{x}}\right)^{25}$ is

(A) $^{25}C_{10}$ (B) $^{25}C_{15}\,(35)^{25}$

(C) $^{25}C_{15}\left(\frac{1}{35}\right)^{35}$ (D) $^{25}C_{10}\left(\frac{1}{25}\right)^{35}$

38. $\int \tan\left(2\tan^{-1}\left(\sqrt{\frac{\sqrt{1+\sqrt{x}}-1}{\sqrt{1+\sqrt{x}}+1}}\right)\right)dx$ equals

(A) $\frac{4}{3}x^{3/4}+c$ (B) $\frac{4}{5}x^{5/4}+c$

(C) $\frac{2}{3}x^{3/4}+c$ (D) $\frac{2}{3}x^{5/4}+c$

39. Find unit vector which is orthogonal to vector $3\hat{i}+2\hat{j}+6\hat{k}$ and is coplanar with vectors $\hat{i}-\hat{j}+\hat{k}$ and $2\hat{i}+\hat{j}+\hat{k}$?

(A) $\frac{2\hat{i}-6\hat{j}+\hat{k}}{\sqrt{41}}$ (B) $\frac{2\hat{i}-3\hat{j}}{\sqrt{13}}$

(C) $\frac{3\hat{j}-\hat{k}}{\sqrt{10}}$ (D) $\frac{4\hat{i}+3\hat{i}-3\hat{k}}{\sqrt{34}}$

40. The curve, which satisfies the differential equation $\frac{xdy-ydx}{xdy+ydx}=y^2\sin(xy)$ and passes through (0, 1), is given by

(A) y (1 - cos xy) + x = 0 (B) sinxy - x = 0

(C) siny + y = 0 (D) cosxy - 2y = 0

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. Let 'z' be a complex number and 'a' be a real parameter such that $z^2 + az + a^2 = 0$, then

(A) locus of z is a pair of straight lines

(B) locus of z is a circle

(C) $\arg(z) = \pm\frac{2\pi}{3}$

(D) |z| =|a|.

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42. If a_i, i = 1, 2, . . ., 9 are perfect odd squares, then

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$$ is always a multiple of

(A) 4 (B) 7
(C) 16 (D) 5

43. Which of the following is /are true ?

(A) $\int_0^1 \sin\left(x^2+2x+1\right)dx - \int_1^2 \sin x^2 dx = 0$

(B) $\int_{-1}^1 e^{\sin x}dx - \int_0^1 e^{\sin(2x+1)}dx = 0$

(C) $\int_{-5}^{-4} \sin\left(x^2-3\right)dx + \int_{-2}^{-1} \sin\left(x^2+12x+33\right)dx = 0$

(D) $\int_{-4}^{4} \cos x^2\, dx - 8\int_0^1 \cos 16(2x-1)^2 dx = 0$

44. Which of the following statement(s) is/are true ?
(A) Number of six digit natural numbers of form abcdef such that a > b > c > d > e > f are $^9C_3$
(B) Number of six-digit natural numbers of form abcdef such that a < b < c < d <e <f are $^{10}C_4$
(C) Number of six-digit natural numbers of form abcdef such that a < b ≤ c < d < e < f are $^{10}C_4$
(D) Number of six-digit natural numbers of form abcdef such that a < b ≤ c < d ≤ e < f are $^{11}C_5$

45. Tangents are drawn from (-2, 0) to $y^2 = 8x$, radius of circle(s) that would touch these tangents and the corresponding chord of contact, can be equal to,

(A) $4\left(\sqrt{2}+1\right)$ (B) $4\left(\sqrt{2}-1\right)$
(C) $8\sqrt{2}$ (D) $4\sqrt{2}$

46. If A and B are non singular matrices of the same order 'n' and A is symmetric matrix, then

(A) $\text{adj}(\text{adj B}) = |B|^{n-2}\, B$ (B) adj A is a symmetric matrix
(C) $\left|\text{adj}(\text{adj B})\right| = |B|^{(n-1)^2}$ (D) $(\text{adj A})^T = (\text{adj A}^T)$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. If $\lim_{x\to 0} \int_0^x \frac{t^2 dt}{(x-\sin x)\sqrt{a+t}} = 1$, then the value of a is

48. If S denote the sum to infinity and S_n the sum of n terms of the series $1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\ldots$ such that $S - S_n < \frac{1}{300}$, then the least value of n is

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49. Find the number of ordered triplets (α, β, γ) satisfying the inequality $2^{\frac{1}{\sin^2\alpha}} 3^{\frac{1}{\sin^2\beta}} 4^{\frac{1}{\sin^2\gamma}} \le 24$ where $\alpha, \beta, \gamma \in (0, 2\pi)$

50. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. If the perimeter of the rectangle is P, then the value of $\frac{P}{20}$ is

51. Let f(x) is a cubic polynomial with leading coefficient unity such that f(a) = b and $f'(a) = f''(a) = 0$. Let $\phi(x) = f(x) - f(a) + (a - xf'(x)) + 3(x-a)^2$ for which conclusion of Rolle's theorem in [a, b] holds at x = 2{a < 2 < b}, then $f''(2)$ is equal to :

52. Let $I_1 = \int_0^x \frac{t^2 dt}{(x - \sin x)\sqrt{\lambda^2 + 2\lambda t}}$ (where λ is non-zero constant) and $I_2 = \int_0^{x^2} \frac{e^t - 1 - t}{x^6} dt$.
If $\lim\limits_{x\to 0} I_1 = 1$, then the sum of squares of value(s) of λ is equal to :

53. If $\sum\limits_{r=0}^{4} \tan\left(\frac{\pi}{15} + \frac{r\pi}{5}\right) = k\sqrt{3}$ then value of k

54. A line makes angles $\alpha, \beta, \gamma, \delta$ with the diagonals of a cube then
$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2 = \frac{\lambda}{3}$ where λ is _____

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